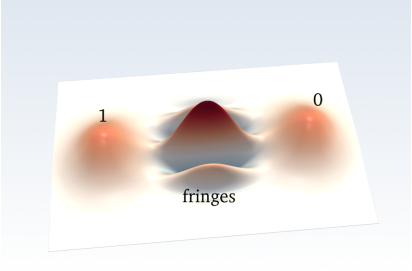
Stabilized cat qubits

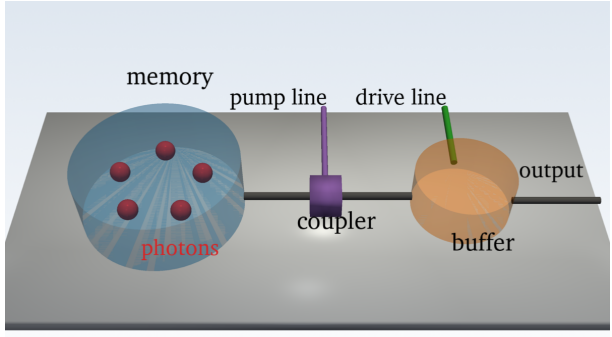
DuQuantum 2026, Alice & Bob challenge

Team Shunya



The device

- Memory resonator stores the qubit, $\kappa_a/2\pi = 9.3$ kHz
- Buffer resonator leaks on purpose, $\kappa_b/2\pi = 2.6$ MHz
- Coupler trades two memory photons for one buffer photon, $g_2/2\pi = 0.763$ MHz

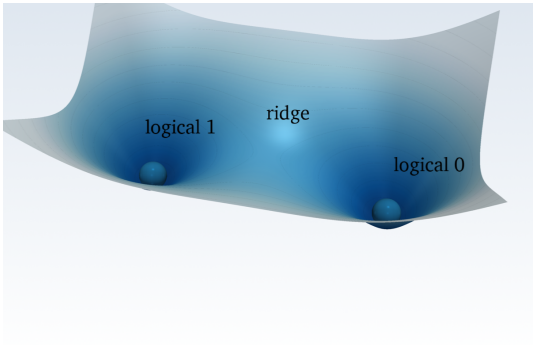


The model

$$H = g_2\left(a^2 b^\dagger + a^{\dagger 2} b\right) - \varepsilon_d\left(b + b^\dagger\right)$$

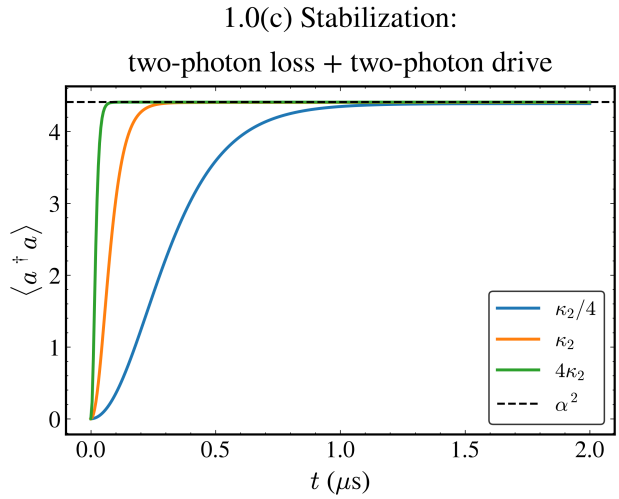
$$L_b = \sqrt{\kappa_b}\, b, \qquad L_a = \sqrt{\kappa_a}\, a$$

- Pairs out and pairs in balance at $\alpha^2 = \varepsilon_d / g_2$
- $\alpha = 2.1$, so 4.41 photons
- Logical 0 and 1 are the valleys at $\pm\alpha$



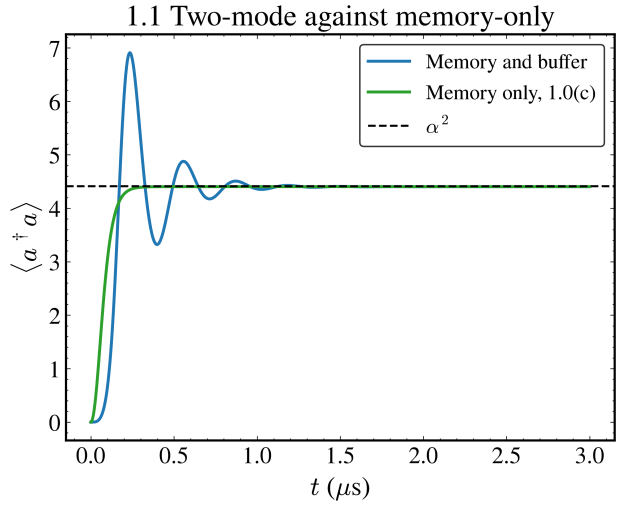
Task 1.0: Warm-up

- One-photon loss matches $4e^{-\kappa t}$ to 6×10^{-12}
- Pair loss stalls at 0.4998, the odd-parity weight
- Stabilization settles at α^2 in a time $\propto 1/\kappa_2$

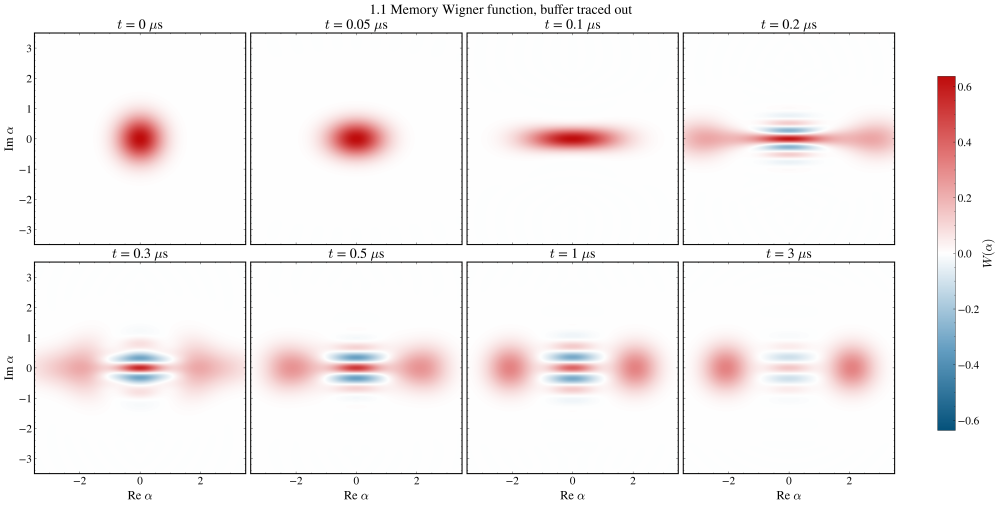


Task 1.1: The full two-mode model

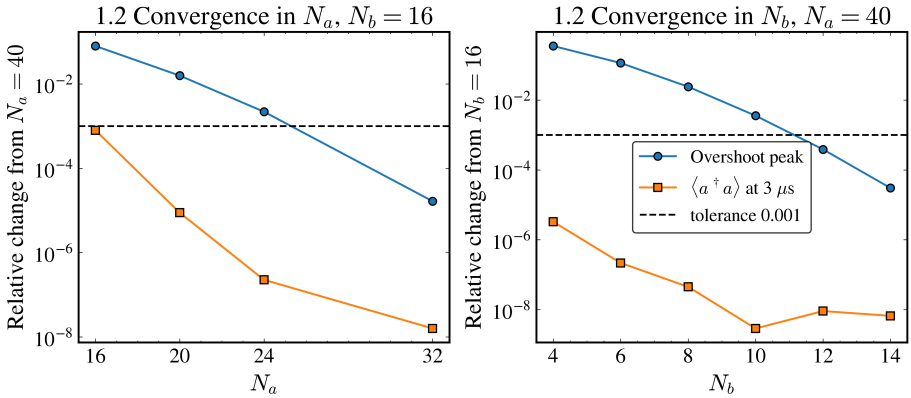
- Memory overshoots to 6.91 photons at 0.235 μs
- Memory and buffer ring with period 0.32 μs, linear theory 0.319 μs
- Memory-only model needs $\kappa_b \gg \varepsilon_d$, here $\varepsilon_d/\kappa_b = 1.29$



Task 1.1: Inflating a cat



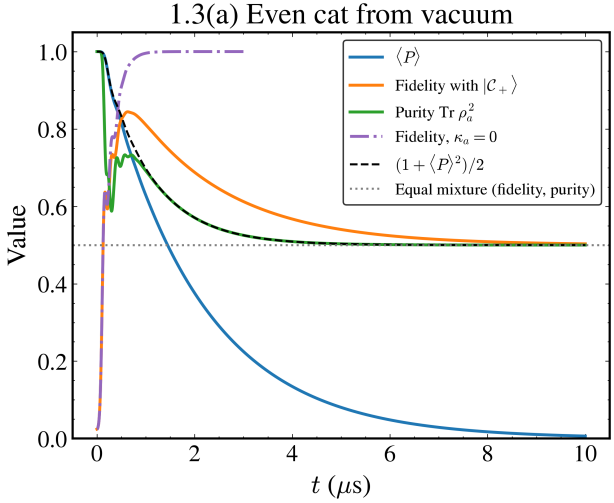
Task 1.2: Convergence study



- Steady state 4.4026, 0.17% below α^2, converged from 16×4 levels
- Overshoot needs 32×12; the suggested $N_b = 6$ is 11.5% low

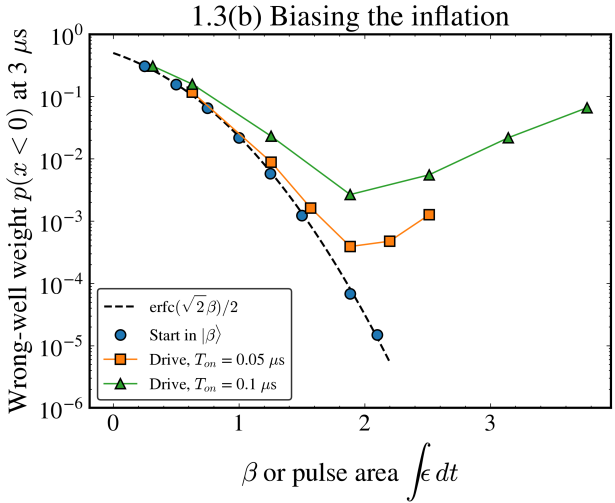
Task 1.3(a): The even cat

- Pair processes conserve parity, and the vacuum is even
- Fringes give $W(0) = (2/\pi)\langle P \rangle$ at every snapshot
- Parity lifetime 1.943 μs; superposition until 1.45 μs



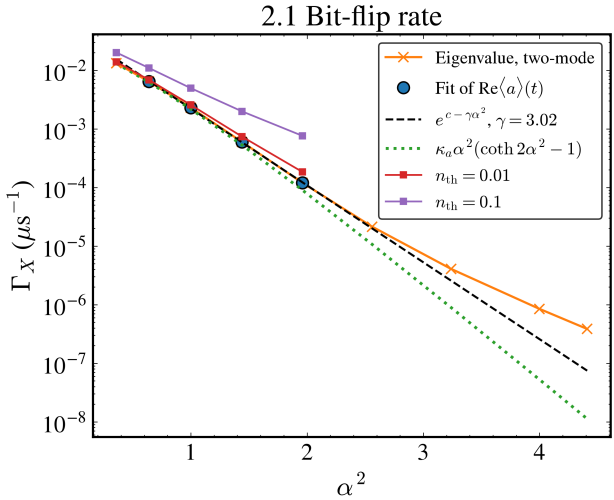
Task 1.3(b, c): Preparing $|+\alpha\rangle$

- 12 MHz, 50 ns memory pulse during inflation: $F = 0.99962$
- Displacing to $\beta = \alpha$ before the pump: $F = 0.999991$
- Cost: drive phase locked to the pump; 30° error gives $1 - F = 3.5 \times 10^{-3}$



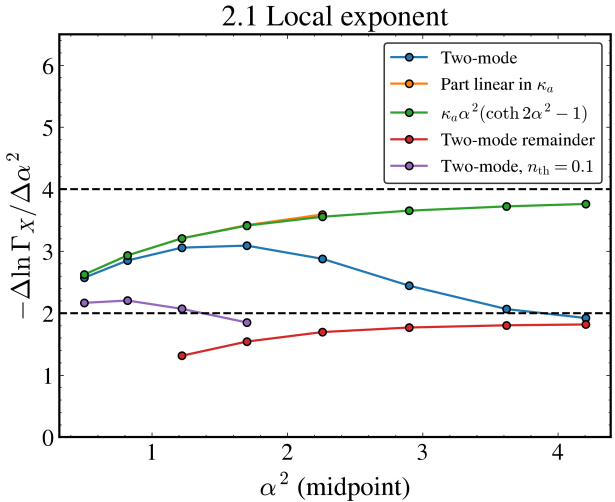
Task 2.1: Bit-flip rate

- $\Gamma_X \propto e^{-\gamma\alpha^2}$, $\gamma = 3.02 \pm 0.04$ for $\alpha = 0.8$ to 1.4
- Time fits and Liouvillian eigenvalues agree to 10^{-11}
- At $\alpha = 2.1$: $T_X = 2.6$ s

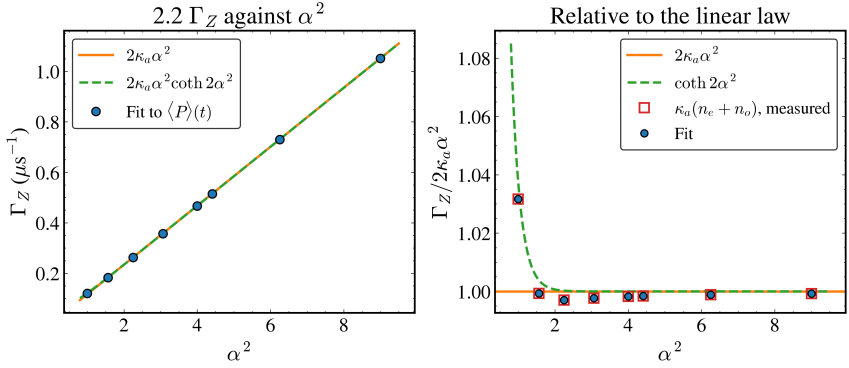


Task 2.1: Why γ is not 2

- Golden rule gives $\gamma = 4$; first-order loss gives 3.22
- A second-order path through the buffer lowers it to 3.02
- That path dominates past $\alpha \approx 1.6$; local $\gamma = 1.9$ at $\alpha = 2.1$
- Thermal photons, $n_{\mathrm{th}} = 0.1$, give $\gamma = 2.02$



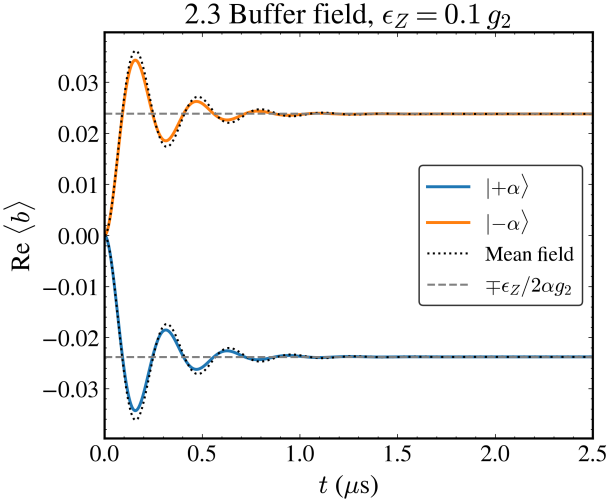
Task 2.2: Phase-flip rate



- $\Gamma_Z = 0.5146\ \mu s^{-1}$ at $\alpha = 2.1$, so $T_Z = 1.94\ \mu s$; slope over $\alpha \geq 1.5$ is $0.99998 \times 2\kappa_a$
- Bias at $\alpha = 2.1$: $T_X/T_Z = 1.3 \times 10^6$

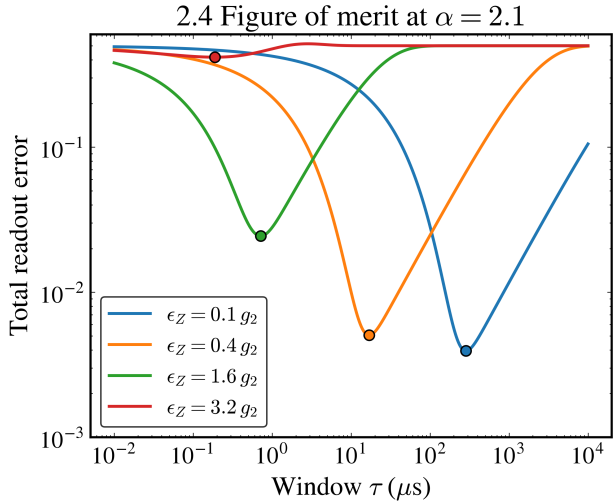
Task 2.3: Readout through the buffer

- Drive $\varepsilon_Z(a+a^\dagger)$ gives $\langle b\rangle = \mp\varepsilon_Z/2\alpha g_2$ to 0.1%
- Rings with period 0.31 μs, settled to 1% after 0.97 μs
- At $\varepsilon_Z = 0.1\,g_2$: 1% error needs 146 μs of homodyne
- A readout with 1% error flips the qubit with probability 1.7 to 1.8×10^{-3}



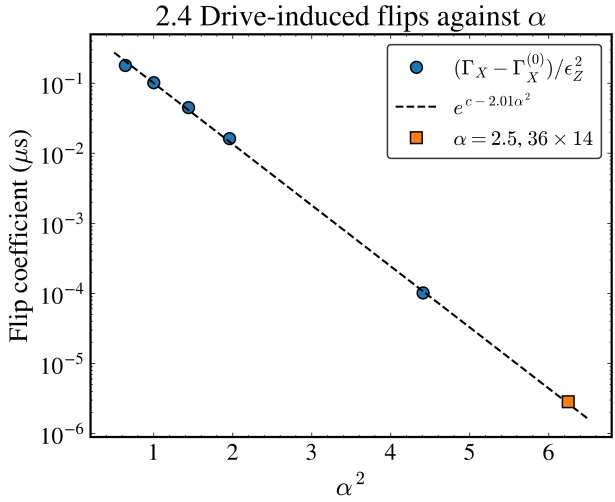
Task 2.4: The readout trade-off

- Total error = misreading + flips caused by the readout drive
- Best at $\alpha = 2.1$: 3.9×10^{-3}, 85% of it from flips
- A stronger drive only trades listening time for flips; above $0.2\,g_2$ it gets worse



Task 2.4: Raise α for the readout

- Drive-induced flips fall about as $e^{-2\alpha^2}$, the signal only as $1/\alpha$
- Reading at $\alpha = 2.5$: error 2.3×10^{-4}, 18 times lower
- Two-step turn-on cuts settling from 0.97 to 0.25 μs
- Circuit level: only the pair term is resonant; the rest are calibratable shifts



Summary

Task	Result
1.1 Inflation	overshoot to 6.91 photons, ringing period 0.32 μs
1.2 Convergence	steady state from 16×4, overshoot from 32×12
1.3 Logical state	even cat; $\lvert + \alpha \rangle$ with $F = 0.99962$ (pulse) or 0.999991
2.1 Bit flips	$\gamma = 3.02$ for $\alpha \leq 1.4$; $T_X = 2.6$ s at $\alpha = 2.1$
2.2 Phase flips	$T_Z = 1.94$ μs at $\alpha = 2.1$, slope $0.99998 \times 2\kappa_a$
2.3 Readout	contrast within 0.1% of theory, settles in 0.97 μs
2.4 Readout optimization	read at $\alpha = 2.5$: error 18 times lower; settles in 0.25 μs

Code, plots and write-up: github.com/yubrajbhandari923/DuQuantum, branch ak-changes